SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Precision (Level 4) **Total Marks: 40**
Question Count: 5

Duration :60 Minutes

Code of Conduct

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform.
It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.
Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professi	Generally clear	Moderately clear	Poorly presented

		onal response			

Total Marks Awarded:

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COURSE CODE: CSA1510

COURSE NAME: Cloud Computing and Big Data Analytics

INDUSTRY PROBLEM SOLVING – BIG DATA SOLUTIONS

1. Retail Data Optimization Problem

1.1 Industry Problem

A retail chain operates through both physical stores and an online shopping platform. It continuously generates large volumes of data from sales transactions, inventory systems, customer browsing behavior, product searches, purchases, and online interactions. Traditional database systems may not efficiently handle this combination of high-volume, high-velocity, and diverse data.

The organization requires a Big Data solution that can provide real-time inventory tracking, accurate demand forecasting, and personalized marketing recommendations. The system should also be scalable because the amount of data will increase as more stores and customers are added.

1.2 Proposed Big Data Architecture

The proposed architecture follows a layered Big Data pipeline:

Data Sources → Data Ingestion → Data Storage → Data Processing → Analytics/ML → Real-Time Applications → Business Users

Data Sources

The system collects data from:

- Point-of-sale (POS) systems
- E-commerce transactions
- Inventory management systems
- Customer browsing and search history
- Mobile applications
- Customer loyalty programs

- Product reviews and feedback

Data Ingestion

Apache Kafka can be used as the real-time data ingestion and streaming platform. It receives sales, inventory updates, and customer activity events continuously.

For batch data, tools such as Apache Sqoop or cloud-based ingestion services can be used to transfer data from relational databases into the Big Data environment.

1.3 Data Storage

A scalable data lake can store structured, semi-structured, and unstructured data.

Technologies such as:

- Hadoop HDFS
- Amazon S3
- Azure Data Lake Storage
- Google Cloud Storage

can be used for distributed storage.

Historical sales records, customer activity, product information, and inventory data can be retained for long-term analysis.

1.4 Data Processing

Apache Spark can process both batch and streaming data.

For example:

- Spark Streaming processes real-time sales and inventory events.
- Spark SQL performs structured data analysis.
- Spark MLlib supports machine learning.
- Batch processing generates historical sales summaries.

A distributed processing architecture allows the retailer to scale when transaction volume increases.

1.5 Demand Forecasting

Machine learning can be used to predict future product demand.

Suitable models include:

- ARIMA for time-series forecasting

- Random Forest
- Gradient Boosting
- LSTM for complex time-series patterns

The model can consider:

Historical Sales + Seasonality + Promotions + Holidays + Price + Location → Predicted Demand

For example, if historical data shows that umbrella sales increase during the monsoon season, the system can forecast higher demand and recommend additional inventory before the season begins.

1.6 Personalized Marketing

A recommendation engine can analyze:

- Products previously purchased
- Browsing history
- Frequently viewed products
- Similar customer behavior
- Product ratings

Collaborative filtering and content-based recommendation techniques can be used to recommend relevant products.

For example, if a customer frequently purchases sports shoes, the system can recommend sports accessories or newly launched running shoes.

1.7 Real-Time Inventory Tracking

When a product is purchased, Kafka can immediately transmit the transaction to the processing layer. The inventory database is updated and the dashboard reflects the new stock level.

A typical flow is:

Customer Purchase → POS/E-commerce → Kafka → Spark Streaming → Inventory Database → Dashboard

Low-stock alerts can automatically be generated when inventory falls below a predefined threshold.

1.8 Security and Risk Awareness

The system should use:

- HTTPS/TLS for data transmission
- Encryption for stored customer data
- Role-Based Access Control (RBAC)
- Authentication and authorization
- Data masking for sensitive customer information
- Audit logs
- Secure API gateways
- Regular backup and disaster recovery

Customer behavior data should only be collected and processed for legitimate business purposes.

1.9 Expected Benefits

The proposed solution provides:

- Real-time inventory visibility
- Reduced stock-outs and overstocking
- Improved demand forecasting
- Personalized customer recommendations
- Faster business decisions
- Scalable data processing
- Better customer experience

Thus, Big Data enables the retail organization to convert large-scale operational and customer data into real-time business intelligence.

2. Social Media Fraud Detection Problem

2.1 Industry Problem

A social media platform handles billions of interactions such as posts, comments, likes, shares, follows, and messages every day. The platform faces two major problems: fake account detection and misinformation detection.

Because new content is generated continuously, traditional batch processing cannot respond quickly enough. A scalable real-time Big Data architecture is therefore required.

2.2 Proposed Architecture

The proposed architecture is:

User Interactions → Kafka → Stream Processing → Feature Extraction → ML Models → Fraud/Misinformation Detection → Alert/Action

2.3 Data Sources

Data can include:

- Posts
- Comments
- Likes
- Shares
- User profiles
- Login patterns
- IP/device information
- Follow relationships
- Account creation activity
- Text and image content
- Reports from users

The data contains both structured and unstructured information.

2.4 Stream Processing

Apache Kafka can collect billions of user events and distribute them across processing nodes.

Apache Flink or Spark Structured Streaming can perform real-time processing.

For example, if an account suddenly sends thousands of comments within a few minutes, the streaming engine can detect the abnormal activity immediately.

2.5 Fake Account Detection

Machine learning can identify suspicious accounts using features such as:

- Account age
- Number of followers
- Number of accounts followed
- Posting frequency
- Login frequency
- Device/IP patterns
- Repeated content
- Abnormal interaction patterns

Models such as:

- Random Forest
- XGBoost
- Logistic Regression
- Neural Networks

can classify accounts as normal or suspicious.

A graph-based approach can also be used because social media relationships form a network.

Users → Followers → Accounts → Interactions

Graph analytics can identify groups of accounts that behave in a coordinated manner.

2.6 Misinformation Detection

Natural Language Processing (NLP) can analyze the text of posts.

The system can perform:

1. Text cleaning
2. Tokenization
3. Feature extraction
4. Sentiment/context analysis
5. Classification
6. Confidence scoring

Transformer-based NLP models can be used for advanced text classification.

The system should not automatically remove every suspicious post. Low-confidence cases can be sent to human moderators.

2.7 Real-Time Detection

A streaming pipeline can work as follows:

Post Created → Kafka → Flink/Spark → Feature Extraction → ML Model → Risk Score → Action

For example:

- Low risk → Normal processing
- Medium risk → Additional verification
- High risk → Alert/moderation review

2.8 Security and Risk Awareness

Important security controls include:

- Encryption
- Secure APIs
- Access control
- Privacy protection
- Anonymization where appropriate
- Model monitoring
- Audit trails

The ML model must also be monitored for false positives and false negatives. Incorrectly identifying genuine users as fraudulent can damage user trust.

2.9 Expected Benefits

The solution provides:

- Near real-time fraud detection
- Scalable processing of billions of events
- Faster misinformation identification
- Automated suspicious-account scoring
- Reduced platform abuse

- Improved user safety

Therefore, combining stream processing, machine learning, NLP, and graph analytics provides an effective solution for large-scale social media fraud detection.

3. Government Data Platform Problem

3.1 Industry Problem

A government agency needs to integrate census, economic, and geospatial datasets collected by different departments. These datasets may use different formats, standards, identifiers, and database technologies.

The major challenge is to create a shared national data platform that provides interoperability while protecting sensitive citizen information.

The platform must support data sharing, analytics, governance, privacy, and security across multiple government agencies.

3.2 Proposed Architecture

The proposed architecture is:

Department Data Sources → Data Ingestion → Data Lake → Data Integration → Governance Layer → Analytics/Data Services → Authorized Departments

3.3 Data Sources

The platform can integrate:

- Census data
- Economic statistics
- Tax-related datasets
- Population information
- Geographic information
- Infrastructure datasets
- Public service data

Both structured and unstructured data should be supported.

3.4 Data Ingestion and Integration

Data can be collected using:

- REST APIs
- Batch uploads
- Kafka
- ETL/ELT pipelines
- Database connectors

Different datasets should be transformed into common formats.

For example, one department may represent a district as District_ID, while another uses DIST_CODE. A master-data management system can map these identifiers into a common standard.

3.5 Storage

A government data lake can use distributed cloud/object storage.

The storage architecture can contain:

- Raw data zone
- Processed data zone
- Curated data zone

Sensitive datasets should have restricted access.

A metadata catalog should record:

- Data owner
- Data source
- Data format
- Data quality
- Data classification
- Update frequency
- Access permissions

3.6 Data Governance Strategy

A strong governance framework is essential.

It should define:

Data Ownership → Data Quality → Metadata → Standards → Access → Privacy → Auditing

Important governance practices include:

- Data classification
- Metadata management
- Data lineage
- Master data management
- Data quality checks
- Retention policies
- Standard data formats
- Interoperability standards

Each department should have clearly defined data ownership and access responsibilities.

3.7 Interoperability

The platform should provide common APIs and standardized schemas.

For example:

Department A → API → National Data Platform → Authorized Department B

Common standards allow different government systems to exchange information without completely replacing their existing applications.

Geospatial data can be managed using spatial databases and GIS technologies.

3.8 Security and Privacy

Government data may contain highly sensitive information. Therefore, security must be a major design requirement.

The system should implement:

- Encryption at rest and in transit
- Role-Based Access Control
- Multi-factor authentication
- Identity and access management
- Data masking
- Data anonymization
- Audit logging
- Network segmentation

- Backup and disaster recovery

Access should follow the principle of least privilege, meaning users receive only the permissions required for their work.

3.9 Expected Benefits

The platform enables:

- Secure cross-department data sharing
- Improved interoperability
- Centralized data governance
- Better policy analysis
- Faster government decision-making
- Reduced duplication of data
- Improved data quality
- Stronger privacy protection

Thus, a combination of Big Data architecture and strong governance provides a secure and interoperable national data platform.

4. Banking Fraud Detection Problem

4.1 Industry Problem

Banks process millions of transactions every minute through ATMs, online banking, mobile applications, credit cards, and payment systems. Fraudulent transactions must be detected within seconds because delayed detection can result in significant financial losses.

The system must combine historical transaction data with real-time transaction streams while maintaining low latency and high accuracy.

4.2 Proposed Big Data Pipeline

The proposed pipeline is:

Transaction Sources → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Score → Decision Engine → Alert/Block

Historical data can simultaneously be processed through a data lake for model training.

4.3 Data Sources

The system can collect:

- Credit/debit card transactions
- ATM transactions
- Online banking transactions
- Mobile banking activity
- Customer profiles
- Device information
- Location information
- Historical fraud records

4.4 Data Ingestion

Apache Kafka can receive transaction events continuously.

Kafka provides distributed, fault-tolerant ingestion and allows multiple consumers to process transaction streams.

For example:

Card Transaction → Kafka Topic → Fraud Detection Service

The transaction does not need to wait for large batch processing.

4.5 Real-Time Processing

Apache Flink or Spark Structured Streaming can analyze transactions in real time.

The system can calculate features such as:

- Transaction amount
- Transaction frequency
- Location change
- Time of transaction
- Device information
- Previous transaction behavior
- Number of transactions within a time window

For example, if a customer makes a transaction in Chennai and five minutes later another transaction occurs from a distant location, the system can assign a higher fraud risk.

4.6 Machine Learning Algorithms

Possible algorithms include:

- Logistic Regression
- Random Forest
- XGBoost
- Neural Networks
- Isolation Forest for anomaly detection

A supervised model can learn from historical transactions labelled as legitimate or fraudulent.

Anomaly detection can identify previously unseen fraud patterns.

The system can generate a fraud probability:

Transaction → Features → ML Model → Fraud Probability → Decision

For example:

- Score < 0.30 → Approve
- 0.30–0.70 → Additional verification
- Score > 0.70 → Block/Review

The exact thresholds should be determined using validation data and business risk requirements.

4.7 Historical Data Processing

Historical transactions can be stored in a data lake and processed using **Apache Spark**.

Historical data is used for:

- Model training
- Fraud pattern discovery
- Customer behavior analysis
- Model validation
- Performance monitoring

The trained model can then be deployed to the real-time detection service.

4.8 Low-Latency Design

To achieve low latency:

- Kafka handles high-throughput event ingestion.
- Flink/Spark Streaming processes events continuously.
- An in-memory feature store can provide frequently required customer features.
- A scalable model-serving service performs predictions.
- Horizontal scaling allows additional processing nodes during transaction peaks.

4.9 Security and Risk Awareness

Banking systems require strong security controls:

- End-to-end encryption
- Multi-factor authentication
- Tokenization of sensitive payment information
- Role-Based Access Control
- Secure APIs
- Continuous monitoring
- Audit logging
- Fraud investigation workflows

The model should also be monitored for **model drift**, because fraud patterns change over time.

A human review process should be available for uncertain cases.

4.10 Expected Benefits

The proposed solution provides:

- Real-time fraud detection
- Low-latency transaction analysis
- High-volume processing
- Improved fraud detection accuracy
- Reduced financial losses
- Continuous learning from historical data
- Scalable banking infrastructure

Therefore, combining streaming analytics, machine learning, distributed processing, and strong security provides an effective solution for modern banking fraud detection.

5. Agriculture Smart Farming Problem

5.1 Industry Problem

Modern agriculture increasingly uses IoT sensors, drones, satellites, weather stations, and farm equipment to collect large amounts of information. This data can be used to improve crop yield, reduce water consumption, optimize fertilizer usage, and identify crop diseases.

However, agricultural data is generated from geographically distributed sources and has different formats and frequencies. A Big Data architecture is required to combine and analyze this information.

5.2 Proposed Big Data Architecture

The proposed architecture is:

Sensors/Drones/Satellites → IoT Gateway → Data Ingestion → Data Lake → Stream/Batch Processing → Analytics/ML → Farmer Dashboard/Automated Actions

5.3 Data Sources

The system collects:

- Soil moisture
- Soil temperature
- Soil pH
- Weather information
- Rainfall
- Humidity
- Crop images
- Drone imagery
- Satellite imagery
- Fertilizer usage
- Irrigation information
- Historical crop yield

IoT sensors continuously generate real-time data.

5.4 Data Ingestion

IoT gateways can collect sensor information from different fields.

MQTT can be used for lightweight IoT communication, while Apache Kafka can handle high-volume event streaming.

The pipeline can be:

Sensor → IoT Gateway → MQTT/Kafka → Processing Platform

This allows data from geographically distributed farms to reach the central platform.

5.5 Data Storage

A cloud-based data lake can store:

- Sensor data
- Historical weather data
- Drone images
- Satellite images
- Crop records
- Yield information

Object storage such as cloud data-lake storage is suitable because it can handle large amounts of structured and unstructured agricultural data.

5.6 Data Processing

Apache Spark can perform large-scale batch processing, while Spark Structured Streaming or Apache Flink can process sensor data in real time.

For example, if soil moisture drops below a threshold, the streaming system can immediately generate an irrigation recommendation.

5.7 Predictive Analytics

Machine learning can predict:

- Crop yield

- Irrigation requirements
- Disease risk
- Fertilizer requirements
- Weather-related crop risks

Models such as:

- Random Forest
- Gradient Boosting
- Regression models
- Time-series models
- Neural Networks

can be applied depending on the prediction task.

For example:

Historical Yield + Soil Data + Weather + Crop Type → ML Model → Expected Yield

5.8 Drone and Image Analytics

Drones can capture images of crop fields. Computer vision models can analyze these images to identify:

- Diseased plants
- Dry regions
- Pest damage
- Weed growth
- Crop stress

A deep learning model such as a CNN can classify or detect unhealthy crop regions.

The system can generate a field health map so farmers can focus resources only on affected areas.

5.9 Precision Irrigation

Real-time soil moisture data can support intelligent irrigation.

Example:

Soil Moisture Low → Streaming Analytics → Irrigation Recommendation → Farmer/Automatic Irrigation System

Instead of watering the entire field equally, irrigation can be targeted to areas that require more water.

This reduces water wastage and improves crop health.

5.10 Security and Risk Awareness

The system should protect:

- Farmer information
- Farm location data
- Sensor data
- Agricultural production records

Security mechanisms should include:

- Device authentication
- Encrypted communication
- Role-Based Access Control
- Secure APIs
- Cloud access controls
- Data backups
- Monitoring and audit logs

IoT devices should also be regularly updated to reduce security vulnerabilities.

5.11 Expected Benefits

The proposed Big Data solution provides:

- Improved crop yield
- Reduced water consumption
- Optimized fertilizer usage
- Early disease detection
- Real-time field monitoring
- Better weather-based decisions
- Reduced operational costs
- Sustainable agricultural practices

Therefore, Big Data, IoT, cloud computing, machine learning, and computer vision can work together to create an intelligent precision agriculture system.